

Response to the reviewers

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Physics-informed e-processes for damage detection under a lifetime false-alarm budget in continuously monitored structures

(revised title; submitted as “Physics-informed e-processes for anytime-valid damage detection in continuously monitored structures”)

Dear Dr. Khudhair and reviewers,

Thank you for the careful and constructive reviews. We have revised the manuscript substantially. In the highlighted version all new or changed text is set in blue; the clean version carries no colour. Section, table, figure and proposition numbers below refer to the revised manuscript. Each comment is summarised in italics, followed by our response and the locations of the changes.

Summary of the main changes

- **Calibration and the guarantee.** The theorem is now stated as a conditional result, and a new part of Proposition 4 shows what any calibration of the bias allowance must control (a calibration-failure term in the error budget) and what it must assume (that the future is represented by commissioning). The implemented rule (commissioning average plus two standard errors) is described as an empirical calibration with no guarantee of its own. A provably calibrated variant (PHASE-C) is evaluated and found to be too conservative to be useful over a one-year window. Abstract, discussion and conclusions are rewritten accordingly.
- **Proofs.** The delay bound (Proposition 5) is re-derived: the inequality direction in the appendix is corrected, the starting-wealth penalty now uses the mixture component to which the drift assumption applies, and the conditioning and stopping assumptions are stated. The identifiability statement (Proposition 1) is generalised and the “if and only if uniform” claim is replaced by a precise statement of when it holds; for the frame studied it does not.
- **New experiments, some unfavourable.** (i) Late damage onset: after two healthy years PHASE detects 5% loss in 66% of records (100% with early onset) and 2% loss in none (Table 9). (ii) A restart variant repairs this but loses empirical false-alarm control. (iii) Re-analysis of Z24 with the recorded order of the data shows that the earlier timing advantage of the cone-restricted process over the omnibus process holds only when commissioning epochs from the two reference days are mixed; we no longer claim it as a general finding (Table 16). The directional result (silence on the stiffness recovery) is unaffected. (iv) The conformal martingale with a full-window calibration set holds its level, which changes our interpretation of that comparator.
- **Reporting.** Exact intervals for all rates, interquartile ranges, explicit definitions of undetected, pre-onset and localisation denominators, a reconciled record count, a corrected statement on PHASE-BB, full parameter listings and a notation list.
- **Claims proportionate to the evidence (second pass).** After the revision was complete we ran a further simulated review and acted on it. The title no longer says “anytime-valid” and now refers to a lifetime false-alarm budget; the abstract is shortened and states that the budget is conditional; Proposition 4(iii) is restated to say that its representativeness assumption is stronger than envelope coverage and is violated by seasonal bias in the simulated generator, so PHASE-C’s zero false alarms are not support for the theorem; arithmetic slips are corrected (the first four record groups sum to 1,800, the healthy monitoring is 1,200 structure-years, the Z24 files hold 700 and 702 epochs); the late-onset table now includes the oracle CUSUM and CTM-F (CUSUM is about eight times faster than PHASE after

two healthy years); the Z24 headings no longer say “confirmed”, R3 is described as a consequence of the sign restriction, and band-edge peak picking, the spalling baseline and the memoryless-chart timings are reported; ablations show that the physics-derived cone adds power beyond a plain sign restriction while withholding identifiable damage from adaptation has no measurable effect over the horizons simulated; a twin-error experiment (10–20% errors) leaves the results unchanged; earlier subspace-based and sequential-test literature is cited and the “under-exploited” wording softened; and the practice section now covers the choice of α , long-lived assets, and the fact that silence is not evidence of safety.

- **Field validation with covariates and masking.** A second real bridge with environmental covariates, KW51 (monitored across a retrofit; hourly frequencies, temperature, humidity, wind), now tests the reduced method (Section 5.4, Table 17, Figure 10). The result is negative: with hourly epochs PHASE alarmed before the retrofit works in 11–13 of 15 commissioning configurations for every surrogate; with daily epochs in 2–7 of 15; only the variant with the certified allowance came near controlling false alarms (0–1 of 15), with delays of two to three months. The configurations overlap, were not pre-specified, and one surrogate was added after the first results, so the table is descriptive. A further simulation (Section 4.2.9) shows that a persistent benign stiffening followed 180 days later by a 5% loss leaves the loss undetected by every cone-based monitor (0/100), while the omnibus channel alarms on the stiffening in 100/100 records at a median of 11 days.
- **Literature and claims.** Direct comparison with the sequential and environmental-compensation studies suggested, a table distinguishing the error criteria of SPRT, CUSUM, e-detectors and conformal martingales, and a narrowed originality claim.

Reviewer 1

Priority 1 / Methodology 1. Establish the connection between calibration (§3.6, commissioning average plus two standard errors) and the conditional score-mean bound assumed by Propositions 3 and 4, including estimation uncertainty, dependence and subsequent environmental variation; otherwise distinguish empirical calibration from the conditional result throughout.

Response. We agree that the commissioning estimate does not establish a uniform bound on future conditional means, and that predictable fitting alone is insufficient. We did two things. First, Proposition 4 now has three parts: (i) the conditional theorem for a known allowance; (ii) for any allowance fixed at the end of commissioning, $\text{FWFA} \leq \alpha + \mathbb{P}(\sigma < \infty)$, where σ is the first epoch at which the allowance is too small, so the calibration-failure probability is part of the error budget; (iii) a finite-sample certified allowance based on the Azuma–Hoeffding inequality, which needs no independence between epochs, together with an explicit *representativeness* assumption (the future conditional excess never exceeds its commissioning average). Estimation uncertainty, serial dependence and later environmental variation enter only through the calibration-failure term and the representativeness assumption, which is the requirement that commissioning spans the operating envelope. Second, we state that the implemented rule has no guarantee of its own: the factor two matches a one-sided error of 2.3% only for independent Gaussian errors, the error is not deducted from the alarm budget, and representativeness cannot be verified from commissioning data. Whether the implemented rule meets the budget is therefore treated as an empirical question. We add a certified variant, PHASE-C, to show the price of a provable statement: no false alarms but almost no power (e.g. 143 of 150 detections of 5% loss at a median of 116 days, none of 2%; Table 6, Table 8). We also examine the sufficient condition directly (Section 4.2.2): the per-epoch bound is not literally met (90-day block means of the excess exceed the allowance in 94.5% of healthy records), yet the discounted process never alarms, so the absence of false alarms must not be read as confirmation of the theorem’s condition. The short-commissioning failures of Table 8 are presented as showing why the implementation’s guarantee must be qualified.

Changes. Section 3.6 and Proposition 4 (new text and proof in Appendix C); Section 3.5 (conditionality sentence); Section 4.2.2 (assumption diagnostics, PHASE-C); Table 4, Table 8; Abstract (Findings); Section 6 (practice, limitations, conclusions: “valid” used only for the conditional results, “calibrated”/“observed” for implemented behaviour).

Priority 2 / Methodology 2. The final inequality in the appendix proof of Proposition 5 does not imply equation (13); the drift assumption concerns one mixture component but the penalty uses total mixture wealth; state the conditioning and stopping assumptions.

Response. The reviewer is right on both counts: the appendix inequality had the wrong direction and the penalty used the wrong quantity. Proposition 5 is restated for a fixed bet $\lambda \in \Lambda$ with $L_t^\lambda = \log \bar{W}_t^\lambda$, the drift assumption on that component, and the bound $\mathbb{E}[(\tau - t^*)^+ \mid \mathcal{F}_{t^*}] \leq (b + B_\lambda + (L_{t^*}^\lambda)^-)/\gamma_\lambda$ with $b = \log(1/\alpha) + \log |\Lambda|$. The proof now uses the mixture-dominates-component inequality, a bounded stopping time $T \wedge n$, optional stopping for the martingale part, the overshoot bound and monotone convergence, in the correct direction. The bound is conditional on $\mathcal{F}_{t^*}$ and applies to $(\tau - t^*)^+$. The corrected penalty turned out to matter in practice: after a long healthy period the high-bet components have lost most of their wealth, which is what the late-onset experiment shows. We also evaluate the bound with plug-in estimates: it held in all 150 records of each of four scenarios, with a median bound about 1.6 times the realised delay (Section 4.2.3).

Changes. Section 3.7, Proposition 5, eq. (13); Appendix C (proof); Section 4.2.3 (bound check); Section 4.2.7.

Priority 2 / Methodology 3. Reconcile Contribution 2 (“iff uniform”) with Proposition 1 (nuisance subspace) and distinguish identifiability relative to nuisance from unique localisation.

Response. Contribution 2 overstated Proposition 1. Proposition 1 now has parts (a)–(d): the confounded set is the subspace $\mathcal{K} = \ker(P_\perp A)$; uniform loss always lies in it; $\mathcal{K} = \text{span}\{\mathbf{1}\}$ holds when $A = D^{-1}S$ has full column rank and the nuisance range contains no direction other than the temperature direction, and fails whenever $\ker A \neq \{0\}$ (in particular when there are more substructures than modes) or when a second nuisance direction lies in the range of A ; and at most $\text{rank}(P_\perp A) \leq m - \dim \mathcal{N}$ combinations of damage can be resolved, so a non-zero identifiable fraction is necessary but not sufficient for localisation. For the eight-storey frame ($n = 8, m = 6, \dim \mathcal{N} = 2$) the confounded subspace has dimension four, and equal loss at storeys 1 and 8 is confounded exactly (cosine -1.000). We recomputed the map for further nuisance directions (Table 7) and replaced the earlier explanation of the storey-1 localisation failure, which the geometry does not support (Section 4.2.5).

Changes. Contribution 2 (Section 1); Section 3.2, Proposition 1 and following text; Appendix C; Section 4.2.5; Table 7; Conclusions.

Priority 3 / Methodology 4. Keep validation within scope; carry Section 5.2’s qualifications into the abstract and conclusions; justify within-state exchangeability; give chronological or dependence-preserving checks; test later onset; or delimit conclusions.

Response. We did both. (a) The abstract, practice, limitations and conclusion sections now state that Z24 supports only a reduced method and cannot separate damage from environment (Table 14 lists, component by component, what each data set validates). (b) Exchangeability is checked and not assumed (Section 5.2): in the ambient records the median lag-one autocorrelation within a state is 0.02 and 3% of series are significant; in the forced records the median is 0.36 and 70% are significant, so exchangeability there is not supported. The 200 orderings are described as a resampling device for one dataset; the reported standard errors are Monte Carlo errors, not sampling uncertainty about structures. (c) We re-ran all Z24 experiments in the recorded order, in reverse, with random circular shifts and with whole measurement set-ups shuffled (Table 16). Directional discrimination is insensitive to the ordering. The earlier timing result is not: it reproduces when set-ups are shuffled across the two reference days (so serial dependence within a

set-up is not the issue) but disappears when the commissioning days are kept apart, because the allowance then rises to 0.58–0.61 from a mean of about 0.06. We therefore withdraw the claim that the cone-restricted process responds sooner to real damage as a general finding. (c') Field data with covariates: the KW51 experiment described above (Section 5.4) is the strongest real-data test we could run with the data available to us; it is a reduced test without a finite element model, and its result is that the reduced method does not reliably control false alarms on a real structure. (d) Late onset: Table 9 places onset at year 3.25. PHASE detects 5% loss in 66 of 100 records (203 days median), none of the 2% losses, and 83 days for 10% (26 days with early onset). A start-time mixture (PHASE-R, Proposition 6) repairs most of the loss but its observed false-alarm probability (0.030 at one year) exceeds the nominal level, because components that start late see only a few months, over which the bias exceeds the commissioning allowance. We report it as a tested remedy with a cost.

Changes. Abstract; Section 5.2 (diagnostics, Table 14, sign-restriction discussion); Section 5.3 (Table 16); Section 5.5; Section 3.8, Proposition 6; Section 4.2.7, Table 9; Section 6.4, Section 6.5.

Priority 4 / Results / Reproducibility. Uncertainty intervals for rates (especially zero-event outcomes); interquartile ranges promised in Table 3; definitions of undetected, pre-onset and localisation denominators; reconcile the 2,700 total; correct the uniqueness claim given PHASE-BB; provide bet grid, clipping, adaptation settings, calibration and comparator procedures and code; precisely describe the fixed-calibration conformal comparator; clarify the 156-day median against the 34% detection rate.

Response. *Intervals:* Table 4, Table 5, Table 9 and Table 8 give counts or exact (Clopper–Pearson) 95% intervals; zero events in 300 records give an upper limit of 1.2% (above the nominal 1%), 0/600 gives 0.6%, and the text says the results are consistent with, and do not prove, the target. *IQRs:* given in Table 5 and Table 9. *Definitions:* Section 4.1.4 defines pre-onset alarms (counted as false alarms, not detections), undetected records (right-censored at the end of the window), delays (medians and IQRs among detected records only, with the unconditional median undefined when fewer than half are detected) and localisation (index read at the alarm epoch; denominator is detected records, stated beside each figure). The 156-day median at 2% is conditional on detection, and the 99 undetected records are censored (Section 4.2.3). (Reading the index at the end of the record over all records, as in the first version, gives the same picture: 89% at 2%, 100% at storeys 3 and 6, 0% at storey 1.) *Total:* the groups in the submission summed to 2,600, not 2,700; Table 2 lists every group, and the revised study has 4,200 records (3,300 two-year, 900 four-year). *Uniqueness:* the claim is removed. PHASE, PHASE-BB, PHASE-E and the full-window conformal martingale all have zero observed false alarms at both horizons and complete detection of losses of at least 5%; Section 4.2.3 says what does distinguish PHASE. *Reproducibility:* Appendix B lists generator, surrogate, adaptation, bet grid (12 values from 0.05 to $0.9/\mu_0$), clipping, allowance, variants and every comparator including calibration procedures; all code and fixed seeds are supplied. *Conformal comparator:* Section 4.1.3 specifies the calibration set, the smoothed p -values and the martingale, and states why validity requires exchangeable, serially independent scores. Because the original calibration set covered only half the commissioning window, we added CTM-F with a whole-window set; it holds the level (0/300), which shows that the excess of the original comparator came from a non-representative calibration set (Section 4.2.2).

Changes. Section 4.1.4; Table 2; Table 4; Table 5; Table 6; Section 4.2.3; Section 4.1.3; Appendix B; `README_reproduction.md`.

Priority 5. Strengthen the comparison with prior work: Kopsaftopoulos and Fassois (2015), Deraemaeker et al. (2008), Maes et al. (2022); distinguish the error criteria of SPRT, CUSUM, e-detectors and conformal martingales; narrow or substantiate the first-application claim; do not generalise the selected baselines.

Response. All three studies are now cited and discussed (Section 2.1, Section 2.3, Section 2.4). We describe what each does, state that PHASE does not compete on environmental compensation, and say that our PCA and chart implementations are single, simple instances calibrated on simulated healthy campaigns whose behaviour must not be generalised. Section 2.3 and Table 1 now

distinguish the assumptions and error criteria of SPRT (error probabilities of one run), CUSUM and Shiryaev–Roberts (average run length, which controls a mean and not a probability within a horizon), e-detectors (average-run-length bound, nonparametric) and conformal test martingales (null of exchangeability of the whole sequence). The first-application claim in the abstract, introduction and research gap is narrowed to “to our knowledge” and to the specific combination of an e-process, a physics-derived cone and identifiability-aware adaptation; we state that earlier sequential tests exist and that we make no wider claim. Bibliographic details are completed (Mammeri et al. article 119862; Shin et al. 1(2):229–260).

Changes. Section 2.1; Section 2.3 and Table 1; Section 2.4; Introduction; Abstract (Originality/value); refs.bib.

Priority 6. Align abstract and practical implications: correct the statement that both limitations reproduce on Z24; identify the 59-day median with its scenario and acknowledge four-year simulations; keep Table 5 illustrative; clarify reset and annualisation; keep contractual claims conditional; notation list and consistent labels.

Response. The abstract is rewritten (see the Abstract responses below). The Z24 sentence is replaced: the commissioning failure mode is confirmed, the identifiability result is confirmed in simulation only, and the abstract states that Z24 cannot separate damage from environment. Table 13 remains illustrative and now states its basis: the 3σ trigger rate is measured under an explicit inspect-and-resume rule (111, 91 and 37 per structure-year for no, one-day and one-week lockouts), replacing the earlier estimate of about 180 per year that was inferred from the median time to a first alarm; a Shewhart chart has no memory, so “reset” changes nothing. Table 13 is restructured so that its rows are commensurable: panel A compares the probability of a first spurious trigger in the first year (1.00, 0.007 and the PHASE budget 0.01), panel B gives the recurrent rate for the 3σ chart only. The PHASE entry is the lifetime budget, not a bound for any calendar year, and the guarantee is stated to concern first alarms, with restarts needing their own budget share; the text says plainly that at $\alpha = 0.01$ PHASE does not promise fewer first-year triggers than the calibrated chart. Contractual and deployment statements are conditional (Section 6.1, Section 6.2). A notation and acronym list is Appendix A; variant labels are PHASE, PHASE-ND, PHASE-C, PHASE-R, PHASE- Ω , PHASE-S, PHASE-NA, PHASE-BB, PHASE-E, CTM and CTM-F throughout (text, tables, figures).

Changes. Abstract; Section 4.2.10, Table 13; Section 6.1; Section 6.2; Appendix A.

Additional questions

Originality.

Narrow or substantiate the first-application claim.

Response. Done as above.

Relationship to literature.

Missing primary studies; decision criteria of SPRT, CUSUM, e-detectors, conformal martingales; bibliographic details.

Response. See Priority 5.

Methodology.

Items 1–4.

Response. See Priorities 1–4; the reproducibility paragraph is answered under Priority 4.

Results.

Intervals for zero-event outcomes; IQRs; denominators; the 156-day median beside 34% detection; PHASE-BB versus the uniqueness claim; the 2,700 accounting; conclusions on the lifetime guarantee and Z24; the

storey-1 localisation claim.

Response. All addressed under Priority 4. For the concluding localisation claim, the conclusions now state that damage was localised correctly at storeys 3 and 6 and never at storey 1 despite a non-zero identifiable fraction, and that identifiability relative to the nuisance is not localisability. We also report that the localisation index has a baseline preference for some storeys on healthy data, so the correct localisation at storey 6 should be read with that in mind (Section 4.2.5).

Practicality.

Contractual commitment depends on conditions calibration has not been shown to enforce; the map is not a localisation guarantee; monitoring evidence at arbitrary times does not extend the guarantee to arbitrary action rules below the threshold; a full calendar year is not a universal requirement; justify reset assumptions and distinguish lifetime allocation from yearly bound; affordability and equity are motivations, not outcomes; specify prerequisites, missing data, recalibration and acceptability of delays.

Response. Section 6.1 now treats the commitment as conditional on commissioning coverage, a coverage test, a restart rule and budget spending across restarts. It states that the map is not a localisation guarantee, that an action rule triggered at evidence level w is protected only at level $1/w$ (Ville’s inequality), and that a full year is what achieved coverage in the simulated seasonal scenario and not a universal requirement. Section 4.2.10 gives the measured trigger rates and the lifetime-average reading of the PHASE entry. Section 6.3 says the sustainability and equity benefits are hypotheses, not demonstrated outcomes, and that the study has no cost data. The new Section 6.2 covers prerequisites, how to judge commissioning coverage, missing data (skipping epochs is harmless only if the decision is independent of the score), recalibration (new process, pre-allocated budget share), and the acceptability of delays (59 days median for 5% loss at one storey; unsuitable for sudden events).

Quality of communication.

Intuitive explanations; notation and acronym list; distinguish c in eq. (7) from c in eq. (9); precise wording for categorical statements; consistency among detecting, attributing and localising; align captions with scenarios (the 59-day median is 5% loss at storey 3).

Response. Short “in words” paragraphs now explain e-processes (Section 3), the damage cone and nuisance subspace (Section 3.2) and the conditional-mean assumption (Section 3). Appendix A is a notation list. The surrogate load coefficient is renamed b_q (temperature coefficient b_T); c is now only the score-saturation scale. Categorical statements about conventional practice are tied to the implementations tested. The 59-day median is identified with the 5% storey-3 scenario wherever it appears.

Abstract.

Reconcile 2,700 two-year records with four-year healthy simulations; clarify that Table 2 uses separate groups of 300; identify the 59-day median; remove the claim that both limitations reproduce on Z24; qualify the lifetime claim; narrow the first-application claim; acknowledge limited detection of smaller damage; simplify terminology.

Response. The abstract now gives 4,200 records (3,300 two-year, 900 four-year); the false-alarm table caption states that the one-year column uses 300 two-year records and the three-year column a separate group of 300 four-year records; the 59-day median is tied to 5% loss at storey 3; the Z24 sentence is replaced as described above; the lifetime claim is conditional; the first-application claim is narrowed; the limited detection of smaller damage (34% at 2%, 66% at 5% after two healthy years) is stated; and the terminology is explained in plain words (a physics-structured surrogate; a damage cone; a running measure of evidence).

Reviewer 2

1. Which components of PHASE are validated by the Z24 data and which only by simulation?

Response. Table 14 answers this component by component. Z24 exercises, in reduced form, the cone (the non-positive orthant), the e-process with the commissioning allowance, and the commissioning-envelope behaviour; KW51 (Section 5.4) adds a reduced test with real covariates (regression surrogate, plain orthant) in which the false-alarm control did not hold; neither data set exercises the physics-structured surrogate (no finite element model, and no covariates in Z24), the nuisance subspace and the identifiability analysis, localisation, or the separation of damage from environment. Section 5.5 states that only the directional result and the dependence on commissioning data survive the additional analyses.

Changes. Section 5.2 (Table 14); Section 5.3; Section 5.5; Abstract; Section 6.4.

2. The cone assumes damage lowers frequencies, whereas the Z24 interventions are settlement, tilt and spalling. Why is the assumption appropriate?

Response. It is a modelling assumption and the data support it only a posteriori. Relative to the reference states, the three tracked frequencies fell in every damage state from 80 mm settlement onwards, in both modalities; only the third mode in the two smallest settlement states rose, by at most 0.6%, comparable to the day-to-day offset (Section 5.2). We do not claim this justifies the assumption in general: mechanisms that raise some frequencies (mass loss, crack closure, boundary stiffening) lie outside the cone and would not be flagged as damage. As implemented, the score is the maximum over singleton, pair and all-mode generators, so a mixed pattern still scores on the falling components. We discuss two mitigations: enlarging the cone with mass-change generators, and running an omnibus process as a separate change-notification channel with its own budget share.

Changes. Section 5.2 (new paragraph on the sign restriction); Section 4.2.4.

3. How would humidity, wind, temperature gradients or changed support conditions affect the identifiability conclusions and the detectability map?

Response. We recomputed the map for a vertical temperature gradient, a support-condition change at storey 1 (equivalently a roof or equipment mass change here) and both (Table 7). Each added direction removes one resolvable dimension ($m - \dim \mathcal{N}$) and lowers the identifiable fractions; a support-condition nuisance makes storeys 1 and 8 entirely confounded and storey 2 nearly so. Effects that act as a uniform modulus or mass change fall inside the existing directions. The map is therefore described as an upper bound that must be recomputed for the nuisance set judged plausible for the asset.

Changes. Section 4.2.5, Table 7; Section 3.2; Section 6.1.

4. The future envelope is unknown in practice. How can sufficient commissioning coverage be determined before relying on the guarantee?

Response. It cannot be certified from commissioning data alone, because representativeness concerns the future (Proposition 4). Section 6.2 proposes a protocol: declare the target envelope from independent records (site climatology, occupancy schedule) and measure coverage; estimate the allowance block by block and treat a large spread as a warning; report coverage online and flag alarms outside the envelope; treat the guarantee as inactive until declared coverage is reached; extend commissioning as a restart with its own budget share; use the twin to synthesise unobserved states and state the model risk. A finer commissioning sweep (Table 8) shows that range coverage is necessary but not sufficient (coverage 97–99% at 180–240 days still left the allowance large and power near zero) and that both coverage and the allowance are needed as diagnostics. Gating alone did not restore validity. None of these steps turns representativeness into a theorem; they make failure visible earlier.

Changes. Section 6.2; Section 4.2.6, Table 8, Figure 7; Section 6.1.

Originality / Methodology / Practicality (summary comments). The complete physics-informed formulation is mainly validated by simulation; transfer of assumptions to Z24 needs justification; further real-structure validation is needed.

Response. We agree and have retained a correspondingly limited claim: the abstract, Section 5.2, Section 5.5, Section 6.4 and Section 6.5 state that the full framework is validated by simulation only and that a campaign with synchronous covariates (Z24 long-term monitoring, KW51) is the necessary next step.

We thank the reviewers and the editor again, and we hope the revised manuscript is suitable for publication.